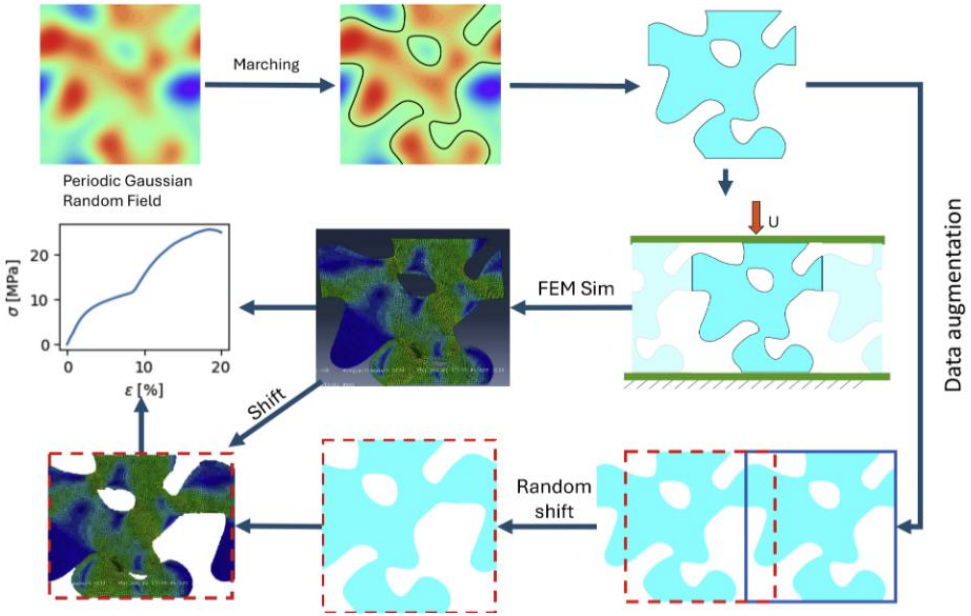
Introduction

- The inverse design of metamaterial unit cells aims to **generate geometries that achieve a target stress–strain response**.



VAR

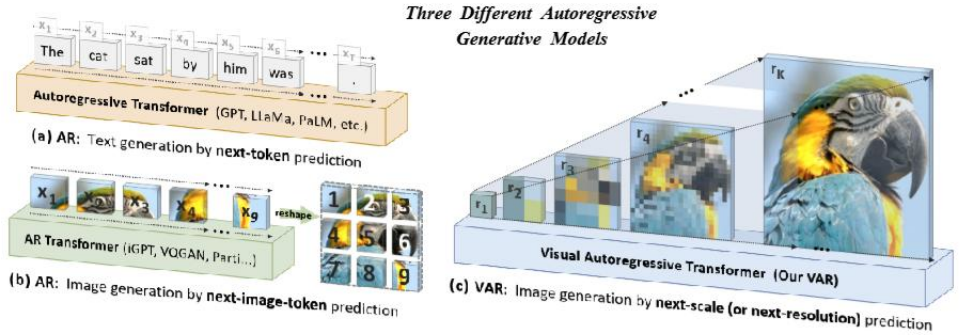


Figure 2: **Standard autoregressive modeling (AR) *vs.* our proposed visual autoregressive modeling (VAR).** (a) AR applied to language: sequential text token generation from left to right, word by word; (b) AR applied to images: sequential visual token generation in a raster-scan order, from left to right, top to bottom; (c) VAR for images: multi-scale token maps are autoregressively generated from coarse to fine scales (lower to higher resolutions), with parallel token generation within each scale. VAR requires a multi-scale VQVAE to work.

$$p(x_1, x_2, \ldots, x_T) = \prod_{t=1}^{T} p(x_t \mid x_1, x_2, \ldots, x_{t-1}). \tag{1}$$

$$f = \mathcal{E}(im), \qquad q = \mathcal{Q}(f), \tag{2}$$

$$q^{(i,j)} = \left(\underset{v \in [V]}{\arg\min} \, \| \text{lookup}(Z, v) - f^{(i,j)} \|_2 \right) \in [V], \tag{3}$$

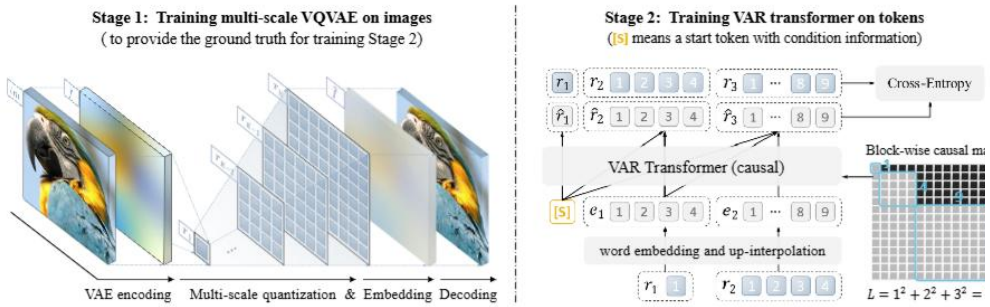


Figure 4: **VAR involves two separated training stages. Stage 1:** a multi-scale VQ autoencoder encod an image into K token maps $R = (r_1, r_2, \ldots, r_K)$ and is trained by a compound loss (5). For details "Multi-scale quantization" and "Embedding", check Algorithm 1 and 2. **Stage 2:** a VAR transformer is train via next-scale prediction (6): it takes $(\texttt{[s]}, r_1, r_2, \ldots, r_{K-1})$ as input to predict $(r_1, r_2, r_3, \ldots, r_K)$. Th attention mask is used in training to ensure each r_k can only attend to $r_{\leq k}$. Standard cross-entropy loss is used

$$p(r_1, r_2, \ldots, r_K) = \prod_{k=1}^{K} p(r_k \mid r_1, r_2, \ldots, r_{k-1}), \tag{6}$$

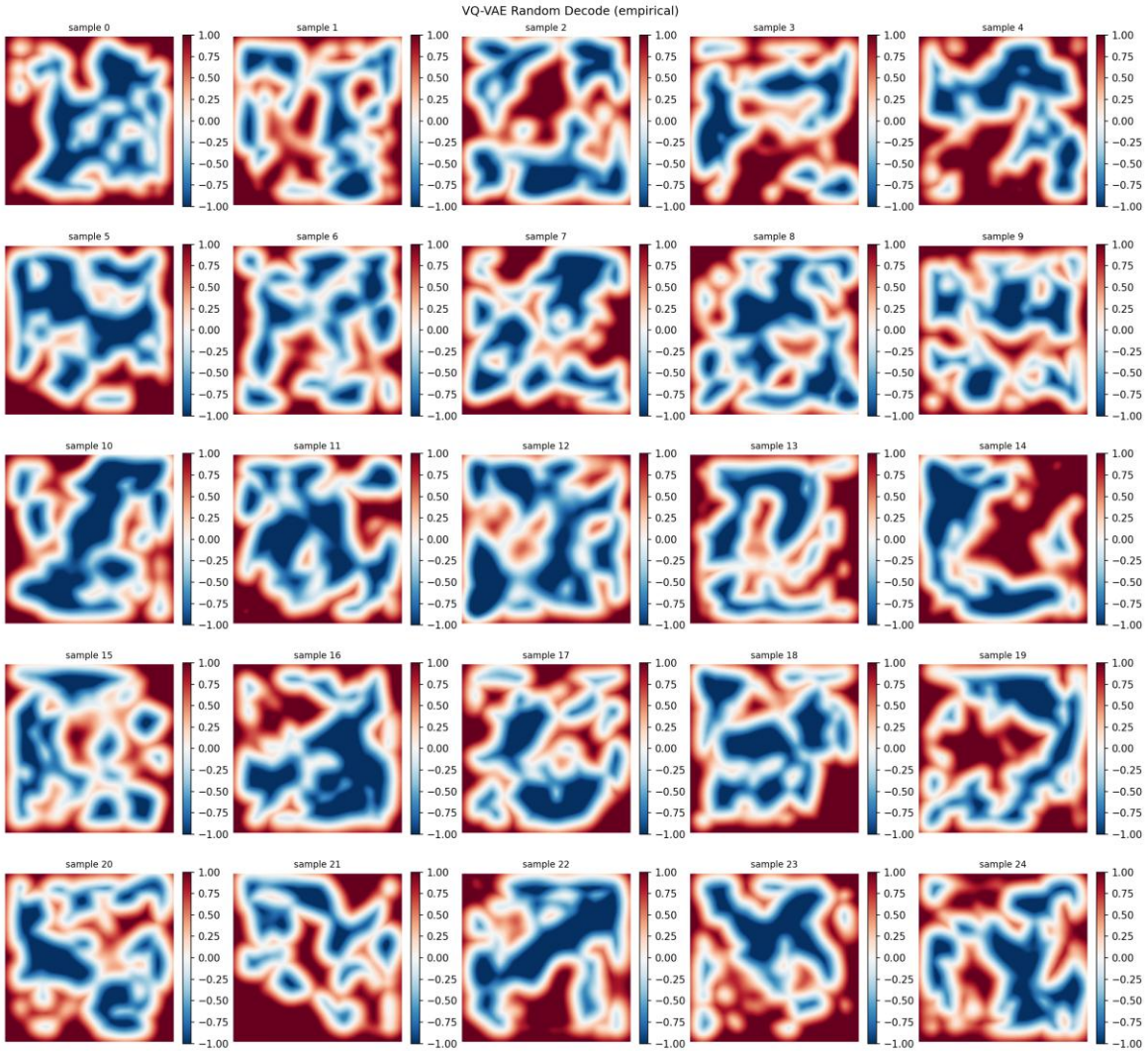
Algorithm 1: Multi-scale VQVAE Encoding

Inputs: raw image im;
Hyperparameters: steps K, resolutions $(h_k, w_k)_{k=1}^{K}$;
$f = \mathcal{E}(im)$, $R = []$;
for $k = 1, \cdots, K$ **do**
  $r_k = \mathcal{Q}(\text{interpolate}(f, h_k, w_k))$;
  $R = \text{queue_push}(R, r_k)$;
  $z_k = \text{lookup}(Z, r_k)$;
  $z_k = \text{interpolate}(z_k, h_K, w_K)$;
  $f = f - \phi_k(z_k)$;
Return: multi-scale tokens R;

Algorithm 2: Multi-scale VQVAE Reconstruction

Inputs: multi-scale token maps R;
Hyperparameters: steps K, resolutions $(h_k, w_k)_{k=1}^{K}$;
$\hat{f} = 0$;
for $k = 1, \cdots, K$ **do**
  $r_k = \text{queue_pop}(R)$;
  $z_k = \text{lookup}(Z, r_k)$;
  $z_k = \text{interpolate}(z_k, h_K, w_K)$;
  $\hat{f} = \hat{f} + \phi_k(z_k)$;
$\hat{im} = \mathcal{D}(\hat{f})$;
Return: reconstructed image $\hat{im}$;

VAR Results


VQ-VAE Random Decode (empirical)
sample 0
sample 1
sample 2
sample 3
sample 4
sample 5
sample 6
sample 7
sample 8
sample 9
sample 10
sample 11
sample 12
sample 13
sample 14
sample 15
sample 16
sample 17
sample 18
sample 19
sample 20
sample 21
sample 22
sample 23
sample 24
1.00
0.75
0.50
0.25
0.00
−0.25
−0.50
−0.75
−1.00